

ω Cen: kinematic data preparation for the dark-matter inference

oCen_dm project notes

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Abstract

This note documents how the proper-motion and line-of-sight dispersion profiles that enter the ω Cen mass inference were assembled: which catalogues, which quality selections, which estimator, and which cross-checks. The central methodological point is that both major proper-motion catalogues contain a large population of stars whose quoted uncertainties are underestimated, and that any dispersion measured from them is a statement about the error model rather than about the cluster. Once each survey’s own quality selection is respected, HST and *Gaia* EDR3 agree in their 300–360'' overlap at $\sigma_{\text{Gaia}}/\sigma_{\text{HST}} = 1.03 \pm 0.13$, the precision being set by the 17 quality-flagged *Gaia* stars available there; a Pristine-based measurement using a different membership definition agrees with *Gaia* to 0.5 ± 1.2 per cent, though it is not an independent astrometric check because its proper motions are the same *Gaia* measurements. Both published profiles are reproduced to a few tenths of a per cent. The final dataset is 127 binned points from five profiles spanning 0.05–57 pc.

1 Scope and principle

The inference asks whether ω Cen requires a dark component beyond its stars and remnants. That question is decided in the outskirts, where a halo would raise the dispersion above what the luminous mass supports. The outskirts are also where the data are worst. Every methodological choice below follows from a single principle, which we arrived at independently and later found stated explicitly by Häberle et al. (2025):

A star whose measurement uncertainty approaches the velocity dispersion being measured makes the inferred dispersion a function of the assumed error model rather than of the cluster.

Such stars are therefore excluded rather than modelled.

2 Data sources

Dataset	Source	Content	Role
oMEGACat VI	Häberle et al. (2025)	1.48×10^6 HST PMs to 466''	primary inner PM
oMEGACat I / MUSE	Häberle et al. (2025)	3×10^5 LOS velocities	inner LOS
<i>Gaia</i> EDR3 members	Vasiliev & Baumgardt (2021)	2.28×10^5 stars to 0.67°	primary outer PM
<i>Gaia</i> DR2 profile	Baumgardt et al. (2019)	9 binned points	outer PM, published
Pristine periphery	Kuzma & Ishigaki (2025)	1.57×10^5 stars to 5.1°	external check
Periphery spectroscopy	Kuzma et al. (2026)	592 LOS velocities to 3.15°	external check

Table 1: Catalogues used. Only the first four enter the likelihood.

3 The dispersion estimator

All profiles measured in this project use the same estimator, applied per annulus.

Model. Each star contributes a two-component mixture. The cluster is a two-dimensional Gaussian whose principal axes follow that star’s own radial direction,

$$\mathbf{C}_i = R(\phi_i) \text{diag}(\sigma_R^2, \sigma_T^2) R(\phi_i)^\top + \mathbf{E}_i + \mathbf{D}_i, \quad (1)$$

with $\mathbf{E}_i$ the star’s proper-motion error covariance and $\mathbf{D}_i$ a rank-one term for the apparent dispersion induced by the cluster’s line-of-sight depth. For *Gaia* and *Pristine*, $\mathbf{E}_i$ carries the catalogue’s correlation coefficient; for HST it is diagonal, because the oMEGACat table supplies no correlation column. Any real HST correlation therefore leaks into the inferred σ_R and σ_T through the omitted $\pm 2\rho\epsilon_\alpha\epsilon_\delta\sin\phi\cos\phi$ terms; the size of that leak is not established here. The field is an *empirical* two-dimensional proper-motion density measured outside the cluster, not a Gaussian. The free parameters per annulus are σ_R , σ_T , the two mean motions, and the field fraction.

Why two dimensions. Projecting the field onto each star’s radial direction mixes its two unequal widths (3.1 and 2.0 mas yr^{−1} for ω Cen) around the annulus and manufactures shoulders that no one-dimensional model reproduces. Fitting in two dimensions removes the problem: the per-annulus χ^2 improved from 8.22 to 1.73 per bin when the field model was changed from two Gaussians to the empirical template.

Error deconvolution. Adding $\mathbf{E}_i$ to the intrinsic covariance *is* the convolution; the fit is unbinned in velocity and predicts the observed distribution directly. No subtraction is performed.

Systemic motion. The cluster’s three-dimensional velocity is projected onto each star’s own tangent basis following Vasiliev et al. (2020, §5.1), which removes the perspective contraction and the rotation of the equatorial basis across the field exactly. Removing nothing at all would leave an azimuthal pattern of amplitude $|\mu_{\text{sys}}| = 7.5 \text{ mas yr}^{-1}$, reading as a spurious dispersion of 5.3 mas yr^{−1}. Subtracting the constant equatorial vector removes that leading term; what the exact projection additionally removes is the residual, of order 0.06 mas yr^{−1} at 2400″.

Rotation. The fitted mean radial and tangential motions absorb streaming, so σ is measured about a rotating mean. The mean-square streaming is retained so that the Jeans model, which predicts the full second moment, is compared like with like. Our fitted mean tangential motion tracks the published rotation curve of Vasiliev & Baumgardt (2021) closely: 0.226 against 0.250 mas yr^{−1} at 435″, 0.159 against 0.177 at 773″, and 0.037 against 0.036 at 1722″.

4 HST

4.1 The catalogue and its quality flag

The oMEGACat VI catalogue contains 1,482,835 sources, 1,385,010 with proper motions, reaching 466″. Its published dispersion profile stops at 300″; the catalogue does not. The flag `selection_hq_astrometry` keeps 48 per cent of the stars with proper motions and requires, simultaneously: a temporal baseline longer than 10 yr; more than 75 per cent of individual measurements surviving clipping; reduced $\chi^2 < 5$ in both components; a proper-motion error inside the lower 95 per cent of the error distribution in its own 0.5-mag bin; and reliable photometry in *both* F625W and F814W. Stars fainter than $m_{\text{F625W}} = 24$ are cut outright, because there the error limit reaches 0.3 mas yr^{−1}, comparable to half the outer dispersion.

The two-filter photometric requirement is not cosmetic: F625W and F814W span 2002 to 2022, so demanding both certifies that the astrometry is supported across the whole baseline.

4.2 Our measurement and its validation

Applying the estimator of §3 to flagged stars only reproduces the published profile across the full radial range:

r (")	N	ours (mas yr ⁻¹)	published	ratio
3.8	146	0.7708 ± 0.0382	0.8048	0.958
8.3	595	0.7944 ± 0.0225	0.7854	1.012
18.0	3224	0.7654 ± 0.0108	0.7776	0.984
39.4	15574	0.7493 ± 0.0106	0.7440	1.007
85.9	66878	0.7070 ± 0.0100	0.7079	0.999
175.8	142615	0.6268 ± 0.0008	0.6236	1.005
270.2	119576	0.5511 ± 0.0008	0.5504	1.001
310.8	16912	0.5257 ± 0.0020	0.5199	1.011

Table 2: Median ratio 1.0042, scatter 2.5 per cent, the scatter arising in the innermost bins where only a few hundred stars pass the flag.

4.3 The outer limit

The flag survives to 360'', on 66 stars in 340–360'' and none beyond. The reason is specific: beyond 360'' no star in the catalogue has F625W or F814W photometry at all, so the two-filter requirement rejects them automatically. This is not a threshold that can be relaxed.

Stars failing the flag are not usable as a substitute. Running the identical estimator on each subset where both exist gives an unflagged-to-flagged dispersion ratio of 1.092, 1.088, 1.079 and 1.052 at 150–200, 200–250, 250–300 and 300–340'', a weighted mean of 1.077 ± 0.011 . This is a ratio of *dispersions*, not of errors: since intrinsic and measurement variances add, an excess of this size would require substantially larger error underestimates, and population differences and residual contamination would have to be excluded before attributing it to errors at all. It is used below only as an empirical sensitivity, never as a calibrated error correction in a fitted dataset. Taken raw, without a field model, they are worse still, and the apparent dispersion *rises* outwards (0.80, 0.88, 1.78 mas yr⁻¹ at 340–380, 380–420, 420–466''), because HST proper motions are relative to the cluster so field stars sit 7.5 mas yr⁻¹ away and dominate a raw variance even at a 1–2 per cent contamination level.

5 Gaia EDR3

5.1 Why the published profile is not used

Vasiliev & Baumgardt (2021) publish a proper-motion dispersion profile on a uniform 24'' grid from 0 to 2400'', obtained as a cubic spline in radius with 2–5 nodes, fitted jointly with a Plummer cluster profile and a two-Gaussian field by MCMC. Five of the eight points our likelihood originally took from it lie inside 380'', at 24, 48, 96, 168 and 336''. Their own quality flag passes *no* star inside 200'' and five between 200 and 300''. The inner points are therefore an inward continuation of a stiff global model, not a measurement. Its quoted uncertainties, as small as 0.002 mas yr⁻¹, are the posterior width of a 2–5 parameter spline constrained by 10⁵ stars, not the precision of any annulus.

5.2 The error model

The released catalogue carries the *raw* Gaia uncertainties. We verified this directly by joining all 228,055 members to `gaia_edr3.gaia_source`: the median ratio of catalogue to raw error is 1.0000,

with 1st and 99th percentiles 0.9992 and 1.0009. The density-dependent inflation $\eta = (1 + \Sigma/\Sigma_0)^\zeta$ with $\zeta = 0.04$, $\Sigma_0 = 10$ (5-parameter) or 5 (6-parameter) is applied inside their fit, not in the file.

Deconvolving the raw errors therefore removes too little noise, and our first outer measurement ran 4–6 per cent high between 460 and 1000". Applying η removes the offset entirely. But η is not correct either: the authors' own non-circular test is that a correct error model makes the deconvolved dispersion independent of magnitude, and it is not. With raw errors the faint-to-bright dispersion ratio is 1.114, 1.135 and 1.077 at 460–700, 700–1000 and 1000–1500"; with η it becomes 0.954, 0.980 and 0.951. Raw errors under-correct, the density-only η over-corrects.

5.3 Our measurement

We therefore avoid depending on either. The profile uses quality-flagged stars with $\frac{1}{2}(\epsilon_\alpha + \epsilon_\delta) < 0.4\sigma_{\text{pub}}(R)$, where σ_{pub} is the published profile used only as a pilot. The intended bound is $\frac{1}{2}(0.4)^2(\eta^2 - 1) \simeq 2$ per cent; it is approximate rather than strict, because the threshold is applied to the arithmetic mean of the two component errors and against a pilot rather than the inferred dispersion, so an individual component may exceed it. The quoted value is the midpoint of the raw-error and η -inflated fits, with half their separation carried as a systematic. That half-separation is a *sensitivity* to the choice of error model; it is not demonstrated to bracket the truth, since both endpoints share the same error-model assumptions.

r (")	N	σ (mas yr ⁻¹)	stat	sys	σ_T/σ_R	f_{field}
356	52	0.4790	0.0335	0.0001	0.949	0.04
437	276	0.4504	0.0141	0.0007	0.823	0.06
533	1477	0.4356	0.0064	0.0017	0.865	0.06
662	2754	0.4019	0.0046	0.0019	0.869	0.10
830	3176	0.3614	0.0040	0.0016	0.942	0.16
1043	2895	0.3204	0.0042	0.0012	0.985	0.34
1328	2552	0.2934	0.0050	0.0008	1.088	0.59
1690	2615	0.2523	0.0070	0.0005	1.081	0.85
2148	3261	0.2314	0.0117	0.0004	1.171	0.95

Table 3: The error-model systematic is 0.1–0.4 per cent against 1.4–6 per cent statistical, so the selection achieved its purpose. The anisotropy is radial inside $\sim 1000''$ and tangential outside, which [Vasiliev & Baumgardt \(2021\)](#) report independently.

The inner edge is 300" because the flag passes five stars in total inside it. The two sparse inner bins are the most error-model-independent in the profile: 53 and 441 stars with errors of 0.025–0.044 mas yr⁻¹ against a dispersion near 0.47, a ratio of 5–9 per cent.

6 Cross-validation

6.1 HST against *Gaia* in their overlap

HST's flag survives to 360" and *Gaia*'s begins at 300", so the overlap is 300–360". Three routes, differing in how HST's rejected stars are used, agree:

Route	HST	<i>Gaia</i>	<i>Gaia</i> /HST
300–340", both flagged	16912 at 311"	17 at 318"	1.028 ± 0.125
300–380", both flagged	16978 at 311"	52 at 356"	0.934 ± 0.065
300–460", HST corrected	92568 at 327"	328 at 430"	0.967 ± 0.029

The three routes are not independent: they share the same HST stars and nest in radius, so they are three treatments of one comparison rather than three confirmations. The first route is

the cleanest: the two samples land at $311''$ and $318''$, essentially the same place, so nothing is extrapolated and no correction is applied on either side. Within an annulus the two samples do not sit at the same radius in general, because HST’s coverage falls outwards while *Gaia*’s flag passes more stars outwards; where that matters both are moved to the annulus midpoint on the local logarithmic slope before the ratio is formed.

A star-by-star match of the two catalogues is possible but informative only about the *unflagged* population, since no quality-flagged *Gaia* star has an HST counterpart. On identical stars, unflagged *Gaia* carries $0.90, 0.79, 0.59$ and 0.44 mas yr^{-1} of scatter beyond its quoted errors at 0–100, 100–200, 200–300 and 300–460'', falling outwards exactly as crowding should.

6.2 A second selection: Pristine

The Pristine periphery catalogue provides a different survey, a different membership definition and a different field model. It is *not* an independent astrometric check: its proper motions are *Gaia* measurements, and the two samples share stars directly, 33 of the 52 selected *Gaia* stars in the innermost annulus and 11–37 per cent further out. A shared astrometric or estimator defect would leave the ratio near unity while both profiles were wrong, and the quoted ratio uncertainty treats the two as independent, which they are not. Selecting on metallicity alone ($[\text{Fe}/\text{H}] < -1.2$; the field median is -0.26 against -1.49 for the cluster) and applying the same estimator with a field template built from the same catalogue beyond 3.6° , with *no* proper-motion cut, gives a ratio to our *Gaia* profile on nine identical annuli from 9 to 57 pc of 1.005 ± 0.012 , with every bin within 5 per cent. Read as a test of the selection and the estimator, not of the astrometry, this is informative: the membership criterion, the magnitude range and the field model all change and the answer does not.

7 Supporting datasets

MUSE. 29 line-of-sight dispersion points from 5.7 to $290''$, taken as published. Line-of-sight and proper-motion dispersions are permitted to differ, since they probe different projections of an anisotropic velocity ellipsoid.

***Gaia* DR2.** Nine published points from 180 to $1747''$ (Baumgardt et al., 2019). Four lie inside $380''$; DR2 has its own, brighter selection and we hold no star list for it, so its usability there cannot be assessed from our side. This remains the least-controlled element of the dataset.

Periphery spectroscopy. 157 flagged members from 0.58 to 3.15° , with no proper-motion selection and therefore free of the circularity that afflicts proper-motion measurements drawn from a proper-motion-selected catalogue. They give $6.12 \pm 0.48 \text{ km s}^{-1}$ at 67 pc and 8.36 ± 1.07 at 83 pc.

8 Beyond the catalogue edge

The *Gaia* member catalogue is truncated at 0.670° (63 pc), which is the authors’ retrieval radius, not a boundary. Computed with AGAMA in the McMillan (2017) potential for $M = 3.55 \times 10^6 M_\odot$ and the orbit integrated from the measured systemic velocity, the Jacobi radius is 90 pc at pericentre and 185 pc at the present galactocentric radius, so the data stop at $0.70 r_J(\text{peri})$. These two numbers are currently hard-coded rather than recomputed by the pipeline, and they use the spherically averaged tidal derivative $\Omega^2 - d^2\Phi/dr^2$; a flattened potential and a different angular-frequency convention would shift them. They are used only to orient the discussion, never in the inference.

A background-subtracted star count in the Pristine catalogue, using eight control windows in proper-motion space, detects the cluster at 78.1 ± 10.2 stars deg^{-2} at 57 pc (7.7σ), 15.9 ± 4.0 at 76 pc (4.0σ) and 2.5 ± 1.1 at 100 pc (2.2σ), falling as r^{-6} and consistent with the field beyond 130 pc in a circular average. The two spectroscopic points that carry the kinematic plateau therefore sit on a real overdensity; the third, at 178 pc from five stars, does not. Beyond 1.6° the 23 surviving candidates are not isotropic (axial Rayleigh $p = 0.016$ at position angle 122° measured from north through east, along-axis excess $0.34 \pm 0.16 \text{ deg}^{-2}$ against -0.02 across), which is what tidal tails look like. Two caveats: the Rayleigh statistic accounts for fitting the axis, but the subsequent wedge comparison is selected on the same data, and the angular uniformity of the survey selection after the metallicity and proper-motion cuts has not been established.

9 The dataset that enters the likelihood

Key	Kind	N	Radial span
hst_pm_radial_ours	PM radial	22	3–341'' (0.08–9.0 pc)
hst_pm_tangential_ours	PM tangential	22	3–341''
muse_los_dispersion	LOS	29	5.7–290''
gaia_dr2_pm	PM combined	9	180–1747''
gaia_edr3_ours_radial	PM radial	9	356–2148''
gaia_edr3_ours_tangential	PM tangential	9	356–2148''

Table 4: 100 points. HST is now our own measurement from flagged stars rather than the published profile, which reaches $360''$ instead of $300''$ and so genuinely overlaps *Gaia*; and *Gaia* enters as its two components rather than their combination, so the anisotropy constrains the model where a halo would act. Fewer points than the previous 127, but each far better measured: the HST uncertainties are 0.1–4.5 per cent against a uniform 1.4 per cent before.

10 Known limitations

1. **The field template does not follow the science selection.** Both the *Gaia* and HST fits score stars against one empirical *Gaia* DR3 field density whose magnitude, quality and error selections differ from the sample being fitted; only its normalisation is free. In the outermost *Gaia* bin, where 95 per cent of the selected stars are field, restricting the template to the same error ceiling moves the fitted dispersion by about 1.5 per cent, well above the quoted error-model systematic. A selection-matched template is the obvious next improvement.
2. **HST’s frame is not restored by adding a constant.** The catalogue’s proper motions are locally corrected, so position-dependent cluster streaming has been removed from them. Adding the constant systemic vector to score them against an absolute field template therefore leaves them displaced where the removed streaming was largest, of order 0.25 mas yr^{-1} . The numerical effect on the membership responsibilities has not been measured.
3. **The error cut uses a pilot profile.** The 0.4σ threshold is evaluated against the published profile, which is the object being replaced. The selection is on errors only and so does not truncate velocities, but its unbiasedness requires the quality selection to be independent of kinematics at fixed radius. ω Cen shows partial energy equipartition (Häberle et al., 2025), so selecting brighter stars selects more massive ones, which have a lower dispersion.
4. **Per-bin errors are treated as independent.** Shared stars between routes, a shared field template, spatially correlated *Gaia* astrometric systematics and a common error calibration are not propagated.

11 Rotation

The Jeans model predicts the full second moment, so each dataset is compared with $\sqrt{\sigma_{\text{model}}^2 - \langle \bar{v}^2 \rangle}$. Where that streaming term comes from differs by instrument, and the difference is not cosmetic: it removes between 1 and 17 per cent of the squared dispersion in the *Gaia* bins.

Gaia supplies its own: our estimator fits a mean motion per annulus. HST cannot. Its proper motions are *locally corrected*, with zero motion defined as the mean motion of the cluster stars in each patch, so the cluster’s streaming has been spent on setting the reference frame. Measured directly, our mixture recovers a residual mean tangential motion of 0.002–0.008 mas yr^{−1} over 40–340″ where the true rotation is 0.054–0.209: 96 per cent of it is gone. HST’s rotation term therefore comes from the Vasiliev & Baumgardt (2021) curve, and our own cannot replace it, because we fit rotation only outside 356″ where HST ends.

The two available *Gaia* curves agree closely (Figure ??, lower right), and swapping one for the other moves the *Gaia* points by 0.07 to 0.49 of their errors, 0.95 σ in quadrature across nine points, with alternating signs. Figures `master_datasets_rotation_ours.png` and `master_datasets_rotation_published.png` show both, and the difference is invisible by eye because the largest shift is 0.30 km s^{−1} on an axis spanning 3 to 26 km s^{−1}. Note that only the *Gaia* points differ between those two figures: HST carries the published curve in both, so the “ours” variant is a hybrid.

12 Open decisions

1. **Decide the status of the four *Gaia* DR2 points inside 380″.** DR2 has its own, brighter selection and we hold no star list for it, so we cannot assess it from here. It is the least-controlled element of the dataset.

References

- Baumgardt, H., Hilker, M., Sollima, A., Bellini, A. 2019, MNRAS, 482, 5138. <https://ui.adsabs.harvard.edu/abs/2019MNRAS.482.5138B>
- Häberle, M., Neumayer, N., Clontz, C., et al. 2025, ApJ, *oMEGACat VI*. <https://arxiv.org/abs/2503.04903>, <https://doi.org/10.3847/1538-4357/adbe67>
- Kuzma, P. B., Ishigaki, M. N. 2025, MNRAS, 537, 2752. <https://arxiv.org/abs/2502.01135>, data <https://doi.org/10.5281/zenodo.14791430>
- Kuzma, P. B., et al. 2026, MNRAS, 550, *Disruption of a Giant*. <https://arxiv.org/abs/2605.23474>
- Vasiliev, E., Baumgardt, H. 2021, MNRAS, 505, 5978. <https://arxiv.org/abs/2102.09568>
- Vasiliev, E., Belokurov, V., Erkal, D. 2020, MNRAS, 501, 2279. <https://arxiv.org/abs/2009.10726>

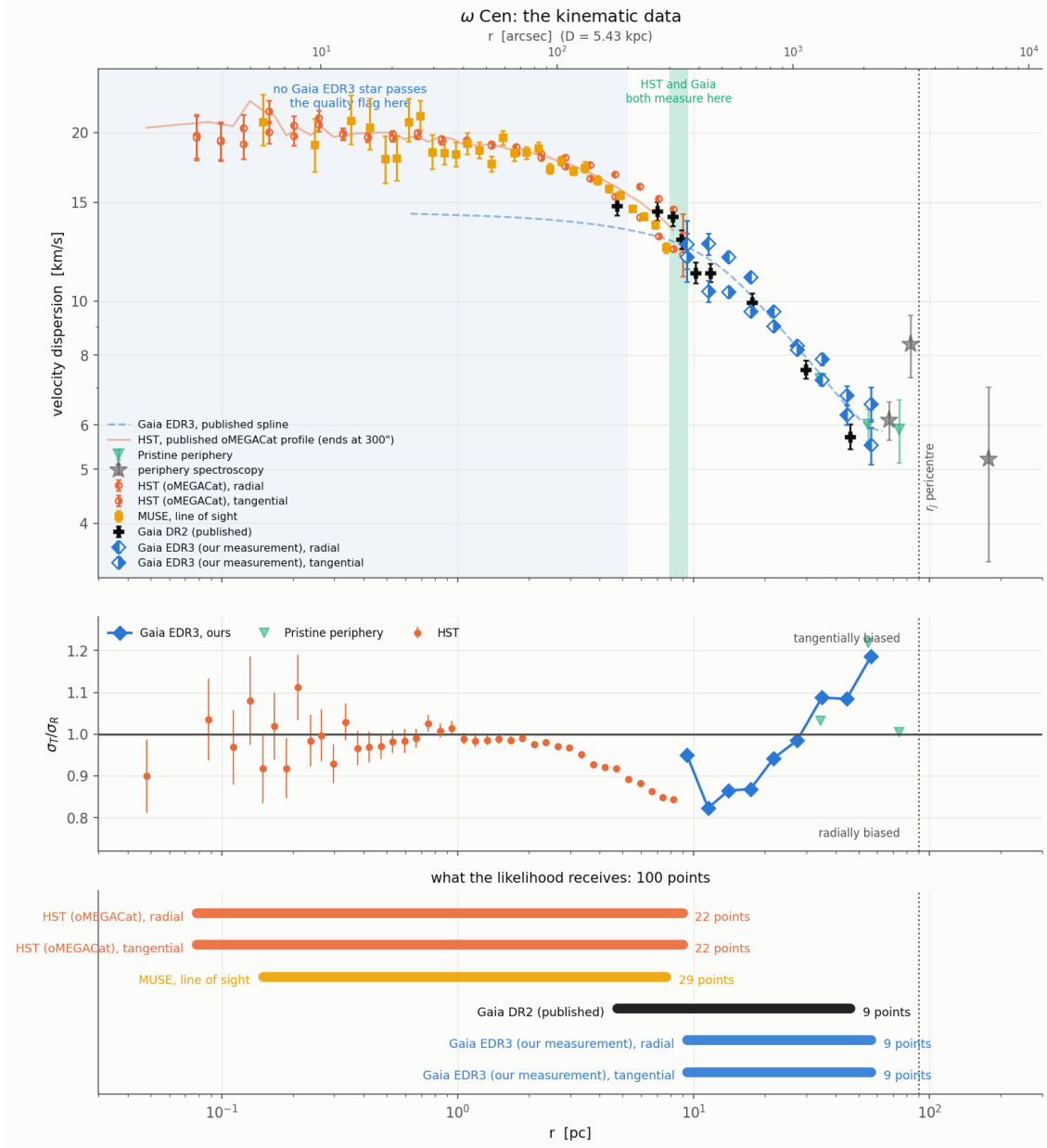


Figure 1: Every dataset in the likelihood, converted to km s^{-1} at $D = 5.43$ kpc so that proper motions and line-of-sight velocities share an axis. One symbol and colour per dataset throughout; the radial and tangential components of the same stars share that symbol and are distinguished by the marker fill. Middle: the anisotropy implied by those components. Bottom: radial coverage and point count.

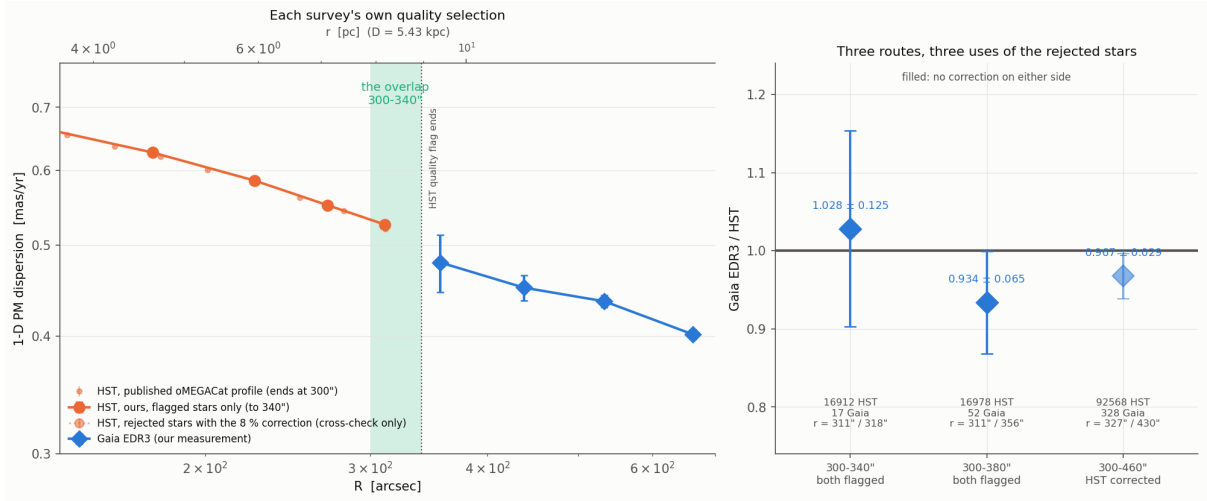


Figure 2: Left: HST measured from its own flagged stars, which reaches $360''$, against *Gaia*, which starts at $300''$. The published oMEGACat profile, which stops at $300''$, is shown faintly. Right: the comparison by three routes differing in their use of HST's rejected stars.

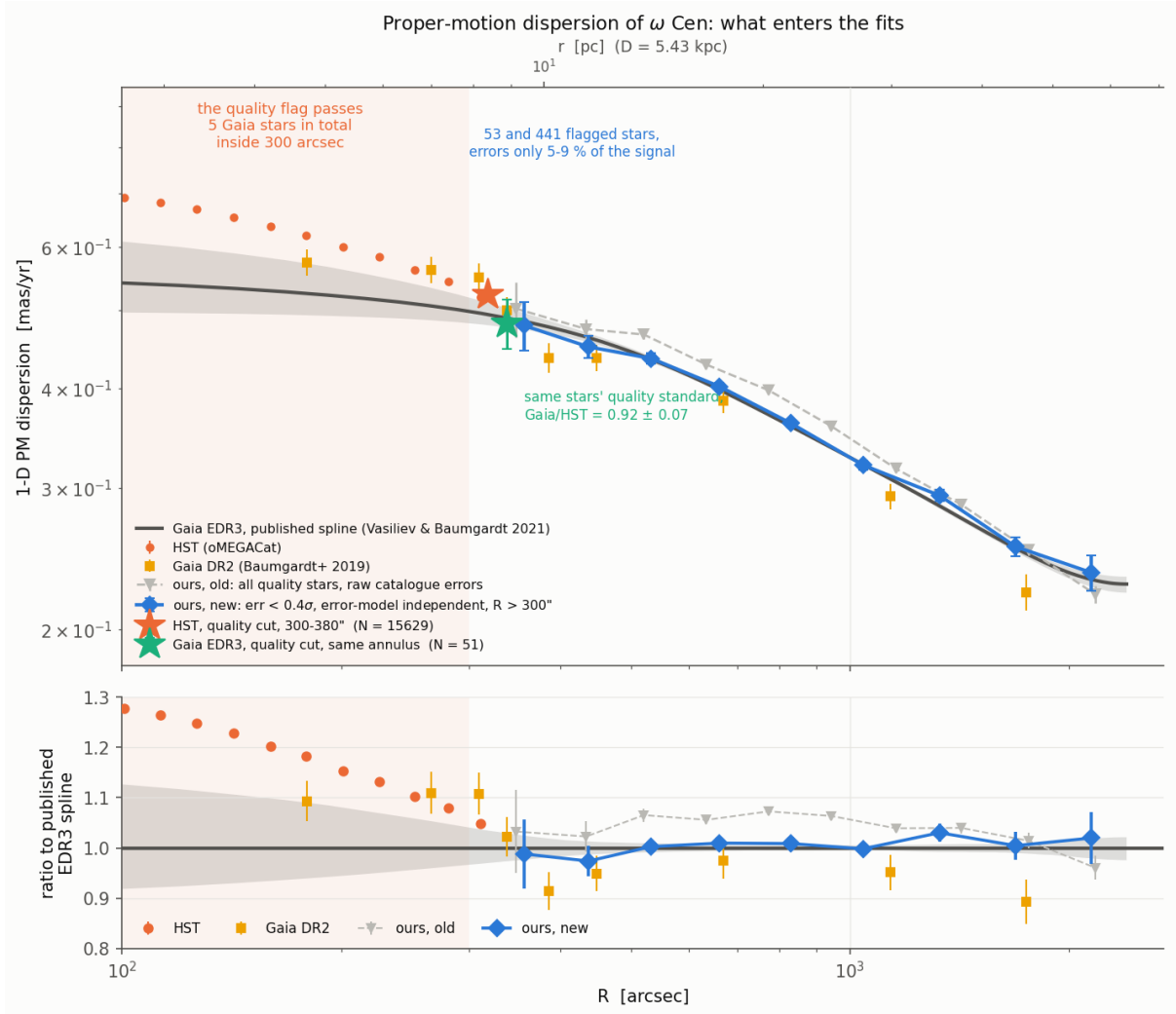


Figure 3: The rebuilt *Gaia* EDR3 measurement against the published spline and against our earlier version, which deconvolved the raw catalogue errors and ran 4-7 per cent high. The lower panel is the ratio to the published spline.

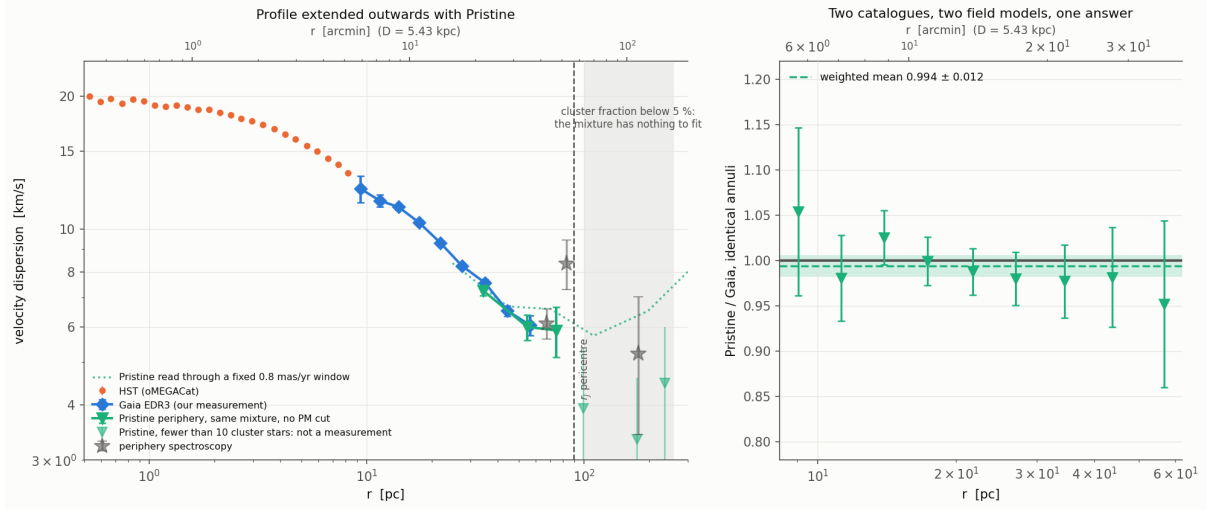


Figure 4: Left: the profile extended outwards with Pristine through the same mixture, with no proper-motion selection, against the same catalogue read through a fixed window. Right: Pristine over *Gaia* on identical annuli, 0.995 ± 0.014 .

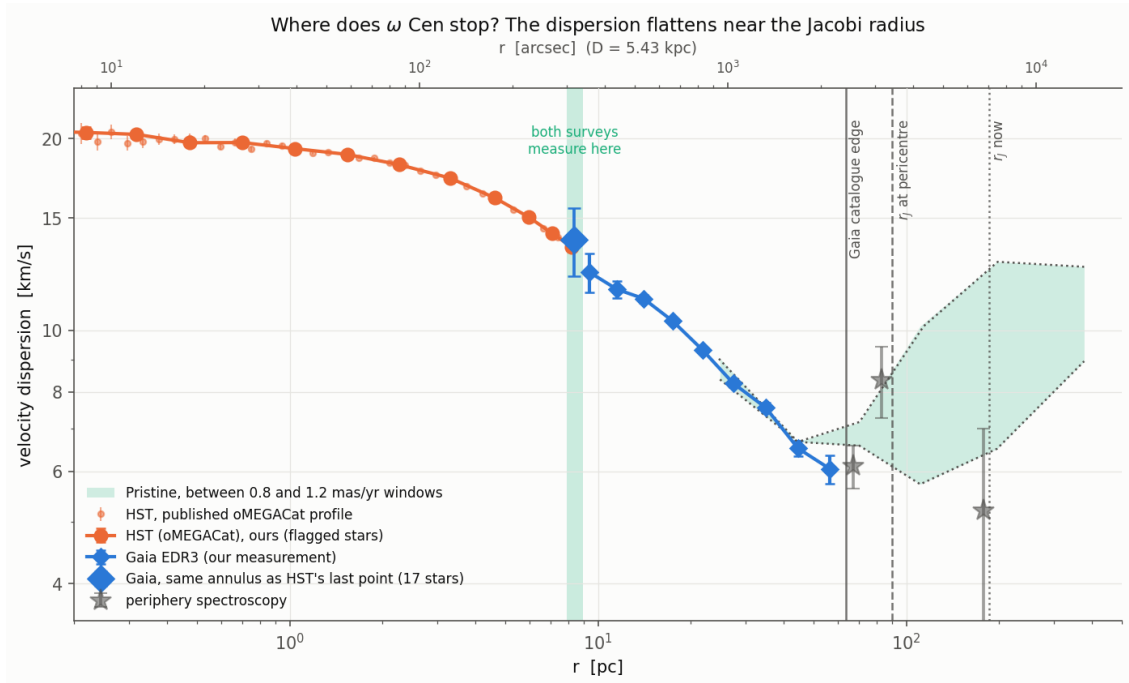


Figure 5: The profile from the centre to 400 pc. The dispersion stops falling near 50–100 pc, where the pericentric Jacobi radius lies.

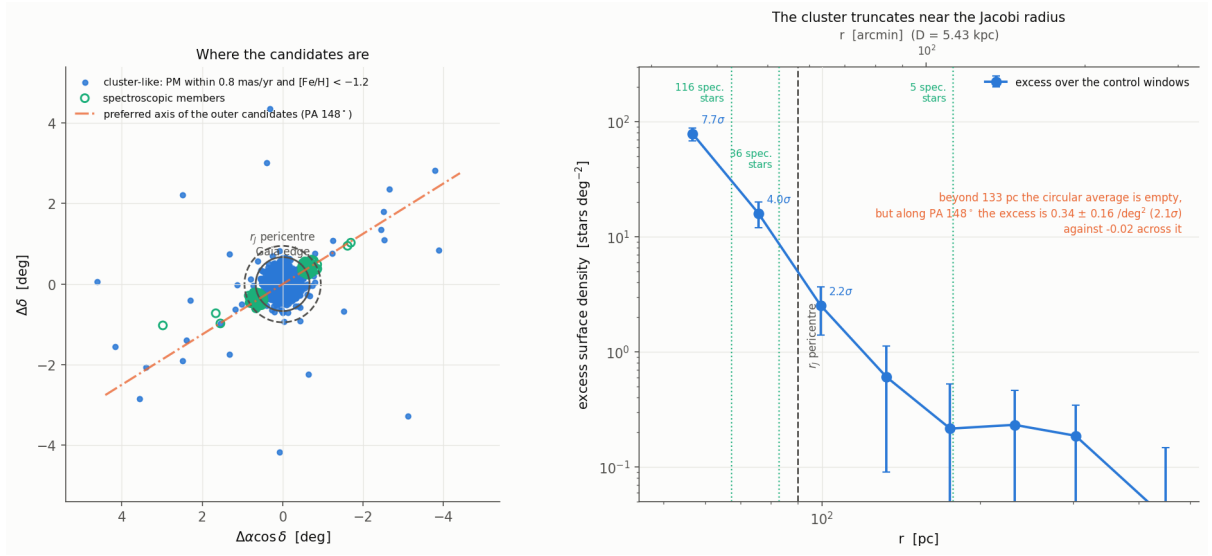


Figure 6: Left: cluster-like stars on the sky, selected on proper motion and CaHK metallicity, with the spectroscopic members overplotted. Right: their background-subtracted surface density, truncating near the Jacobi radius.